Multiobjective Optimization for Common-Centroid Placement of Analog Transistors

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***Abstract*—In analog circuits, process variation can cause unpredictability in circuit performance. Common-centroid (CC) type layouts have been shown to mitigate process-induced variations and are widely used to match circuit elements. Nevertheless, selecting the most suitable CC topology necessitates careful consideration of important layout constraints. Manual handling of these constraints becomes challenging, especially with large size problems. State-of-the-art CC placement methods lack an optimization framework to handle important layout constraints collectively. They also require manual efforts and consequently, the solutions can be suboptimal. To address this, we propose a unified framework based on multiobjective optimization for CC placement of analog transistors. Our method handles various constraints, including degree of dispersion, routing complexity, diffusion sharing, and layout dependent effects. The multiobjective optimization provides better handling of the objectives when compared to single-objective optimization. Moreover, compared to existing methods, our method explores more CC topologies. Post-layout simulation results show better performance compared to state-of-the-art techniques in generating CC layouts.**

***Index Terms*—Analog circuit, common-centroid layout, diffusion break, dummy count, layout-dependent effects, mismatch, multiobjective optimization, placement, routing.**

I. Introduction

IN ANALOG circuits, process variation can affect matching of the devices, which degrades circuit performance [1]. Process-induced variations can be categorized as random variations and systematic variations. One effective technique to reduce random variations is to make the devices bigger [1], [2]. However, increasing the size may cause the devices to become more sensitive to process gradients [2]. Common-centroid (CC) layouts have been shown to be better than other alternatives such as clustered and interdigitated patterns for mitigating process-induced linear variations [3] and are widely used to match circuit elements [2], [4], [5], [6], [7]. In a CC layout, each device is broken into multiple units and the units are placed in different locations in an array in such a way that the spatial centroids of the devices overlap. A CC layout is symmetric about both the X and Y-axes.

However, the challenge in implementing a CC type layout comes from the fact that there can be many CC topology configurations, i.e., the devices can be placed in many different ways for the centroids to overlap. Moreover, one must consider various layout constraints while optimizing a CC layout. These include maximizing degree of dispersion to achieve uniform device spread, a factor affecting variation performance [2], minimizing route length to reduce parasitics and voltage drop, maximizing diffusion sharing to reduce layout area, and minimizing layout dependent effects (LDEs) such as length of diffusion (LOD) and well proximity effects (WPEs) to mitigate threshold voltage change. Considering these competing constraints, manually selecting the optimal CC topology becomes challenging [8], [9].

Over the last decade, various studies have delved into CC placement of analog devices including transistors, capacitors and resistors. Some earlier approaches have focused on generating high-quality CC topologies, mainly considering spatial variation, not layout effects [2], [10], [11], [12], [13]. On the other hand, methods proposed in [4], [11], [14], [15], [16], [17], and [18] may not apply to general transistor circuits. The recent works presented in [3], [9], [19], and [20] have made substantial progress in incorporating layout constraints into CC placement. However, these constraints are addressed separately through post-processing steps following CC topology generation. For instance, to enhance circuit offset performance, dummy components are placed around the CC structure, effects of parasitics and electromigration are taken into account during routing phase following the placement step [19], [21]. One significant drawback is the lack of an optimization framework capable of collectively addressing the constraints, which can lead to suboptimal results. The work in [21] introduces a simulated annealing-based optimization framework that handles several layout constraints while optimizing nonlinear spatial variation. However, the final layout type achieved using this method is not CC. Unlike [3], [9], [19], where the fixed nature of the formulation limits exploring CC topologies, the simulated annealing-based approach in [21] allows exploration of various non-CC topologies. While [3], [9], [19] present post-layout simulation results, the experimental findings in [21] are model-based.

We introduce a unified multiobjective optimization framework to generate CC-type layout of analog transistors while handling important layout constraints. Unlike single-objective optimization method, which requires careful tuning

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of coefficients to balance different objectives, multiobjective approach eliminates the need for tuning. Specifically, we enhance a well-known multiobjective optimization algorithm AMOSA [22]. AMOSA has been used for solving circuit-level placement problem [23]. Here, we use it to address device-level placement problem. AMOSA relies on the concept of the amount of domination rather than coefficient-based control of objectives. The use of an archive to store nondominating solutions seen during the optimization process allows diverse exploration of solution space. Moreover, compared to [3] and [19], we explore more CC topologies by applying powerful transformations.

The key contributions of our work are as follows.

1) To the best of our knowledge, we are the first to propose an optimization framework for CC placement of analog transistors.
2) We enhance a well-known multiobjective optimization algorithm, AMOSA, to handle more than one new solution per iteration.
3) Furthermore, we consolidate several cases/subcases of the AMOSA algorithm into just three cases, simplifying the algorithm and making the code easier to implement.
4) We explore significantly more CC topologies compared to the state-of-the-art by applying powerful transformations.
5) Our optimization formulation encompasses important layout constraints, including diffusion break, LDEs, routing cost, and degree of dispersion.
6) Post-layout simulation results show that the proposed method performs better than state-of-the-art across different circuit configurations and important constraints.

The rest of this article is organized as follows. Section II discusses our CC placement optimization framework. Section III presents various layout constraints handled by our framework. Results are discussed in Section IV. Section V concludes this article.

II. CC Placement Optimization

Simulated annealing, a classical optimization technique [24], has been used for both digital [25], [26] and analog circuit placement [14], [21], [27], [28]. The main idea is to optimize circuit performance by perturbing potential placement solutions with actions such as random selection, swapping, and rotation. However, the single objective optimization technique in these approaches hinges on a cost function. The selection of coefficient values for various objectives in the cost function is a manual task. In [22], an algorithm AMOSA for multiobjective optimization based on simulated annealing has been proposed. AMOSA uses a concept of amount of domination to compute the acceptance probability of a new solution. It utilizes an archive to retain the nondominated solutions encountered so far. To better understand the concept, consider two solutions, denoted as sol_1 and sol_2. sol_1 dominates sol_2, if $\forall i \in \{1, 2, \ldots, M\}$, $f_i(\text{sol}_1) \leq f_i(\text{sol}_2)$, where f is the objective to be minimized and M is the number of objectives. For the two solutions, the amount of domination is defined as follows:

$$\Delta\text{dom}_{\text{sol}_1,\text{sol}_2} = \prod_{i=1, f_i(\text{sol}_1)\neq f_i(\text{sol}_2)}^{M} \frac{|f_i(\text{sol}_1) - f_i(\text{sol}_2)|}{R_i}. \quad (1)$$

Here, R_i represents the range of the ith objective. The algorithm begins by entering an initial solution, termed cur-pt, into the archive at temperature $T_{\max}$. The cur-pt is perturbed to yield a new solution, referred to as new-pt. The domination status of the new-pt is then checked with respect to the cur-pt and the solutions within the archive. The archive and the cur-pt are updated based on the domination status. This process iterates a total of n times for each temperature. The temperature is reduced to α x temp, using the cooling rate α, until the minimum temperature, $T_{\min}$, is reached. Upon reaching $T_{\min}$, the iteration concludes, and the archive contains the final nondominated solutions. Post-processing can be applied to this archive to get the most desired solution.

A. Initial Placement With Min. Diffusion Break

Diffusion sharing is a widely used concept in analog applications [19]. Sharing diffusion region not only reduces layout area and routing cost but also minimizes spatial variation [21]. However, fully sharing diffusion region without using dummies is not always possible in a CC type layout, as diffusion breaks could be unavoidable due to other constraints [3], [4], [19], [21]. We present two cases in Fig. 1(a): one where diffusion break can be avoided and the other where it is likely to occur, necessitating the use of a dummy transistor. A lower value of diffusion break does not always translate to a smaller overall area. The location of the diffusion break can impact the need for additional dummy transistors to maintain a CC structure while sharing diffusion region. For example, shown in Fig. 1(b), compared to solution 2, solution 1 requires more layout area and routing resources, although it has fewer diffusion breaks.

However, our findings indicate that minimizing only the dummy count leads to inferior results. We, therefore, optimize both the dummy count and the diffusion break. First, we generate an initial placement that has a minimum number of diffusion breaks. Subsequently, during optimization, a new solution is accepted only if the diffusion break and the number of dummies do not increase beyond an upper bound, which can be a user-defined constraint. We create the initial CC placement (i.e., the cur-pt) by placing half of the devices and mirroring the other half (XX/180° transformation, where X is a device [2]). For placing half the devices, we adopt a sequential placement approach, grouping units of the same device together. Consecutive devices share drain or source terminal. This is illustrated in Fig. 2. To generate the new solution (i.e., the new-pt), we randomly swap two distinct unit devices within half of the device set and mirror this action in the other half.

B. CC Topology Space Exploration

However, the mirroring strategy of device placement using the XX/180° transformation proves insufficient. To explore

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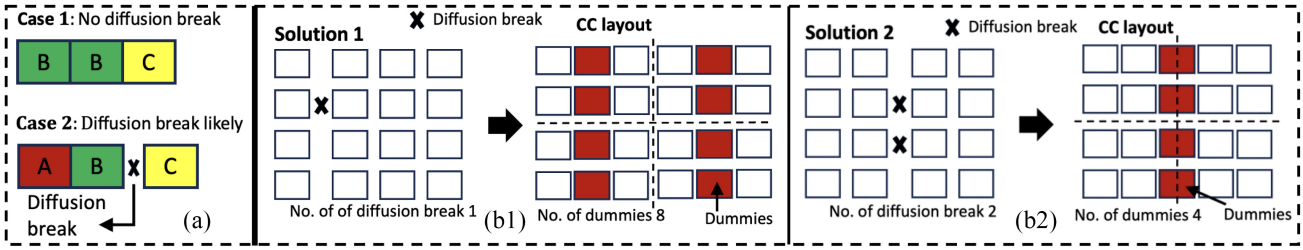


Fig. 1. (a) Case 1: No diffusion break, Case 2: Diffusion break likely. Here, A, B and C are unit transistors sharing drain or source terminal. (b1) and (b2) Solution 1 has one diffusion break compared to two in solution 2, however, solution 1 requires eight dummy insertions compared to four in solution 2 to maintain CC structure while sharing diffusion region.

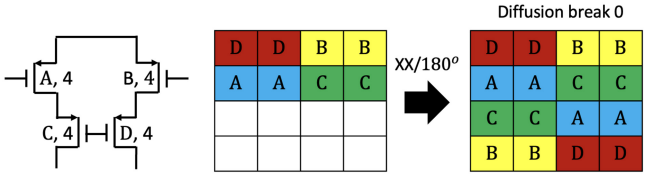


Fig. 2. Generating initial CC placement that has minimum number of diffusion break for the circuit on the left by applying XX/180° (where, X ∈ {A, B, C, D}) transformation on half of the devices placed sequentially.

more CC topologies, we also employ XY/180° transformation, where X and Y are devices [2]. The results of these two transformations for a two-transistor layout with four units each yield six distinct layout types, four of which adhere to the CC configuration, as shown in Fig. 3. We have not shown the XX/180° or XY/180° transformation on "B A A B" or "A B A B" as they would not produce any new topology or distinct pattern. Note that the method proposed in [19] does not perform a topology search and is thus restricted to generating a single CC topology. While there are many other transformations possible through rotational and reflectional symmetries, the XX/180° or XY/180° transformations are considered the most powerful [2]. These transformations are applied at the perturbation stage in each iteration of the multiobjective optimization algorithm run. The handling of transformation in a general case of perturbation that includes random swap is illustrated in Fig. 4. Note that XY/180° transformation can be applied only to devices having the same number of units. If a transformation yields a non-CC type layout, it is not considered a solution.

C. Enhancement to AMOSA

We enhance the original AMOSA algorithm [22] to handle a specific situation in our problem where the perturbation stage may generate more than one new solution. Note that the XX/180° transformation yields only one solution, while the XY/180° transformation can generate multiple solutions as each pair of devices with the same number of units can produce a unique topology. For instance, in Fig. 4, each combination <A B>, <A C>, <A D>, <B C>, <B D>, <C D> produces a distinct topology, although only <B C> qualifies as CC. Unlike AMOSA, which generates one new solution in each iteration and stores it in new-pt, we store the solutions in new-pts. Subsequently, we have the task of updating the cur-pt and the archive. We consider three cases for updating the cur-pt. *Case 1:* cur-pt dominates k_1 (> 0) solutions in new-pts, *Case 2:* cur-pt nondominates all solutions in new-pts and *Case 3:* k_1 (> 0) solutions in new-pts dominate cur-pt. At any iteration, new-pts and archive contain only nondominated solutions.

1) *Case 1:* Given that k_1 solutions out of a total k solutions in new-pts are dominated by cur-pt, the remaining $k-k_1$ solutions are, by transitivity, nondominating. Therefore, we randomly choose one solution from $k-k_1$ solutions and assign it to cur-pt with some probability. The probability (*prob*) calculation considers the degree of domination of k_1 solutions in new-pts by cur-pt and the solutions in the archive. We modify (2) in [22] as follows:

$$\text{prob} = \frac{1}{1+\exp\left(\frac{\Delta\text{dom}_{\text{avg}}}{\text{temp}}\right)}. \tag{2}$$

where

$$\Delta\text{dom}_{\text{avg}} = \frac{\left(\sum_{i=1}^{k_3}\sum_{j=1}^{k_2}\Delta\text{dom}_{i,j}\right)+\sum_{i=1}^{k_1}\Delta\text{dom}_{\text{cur-pt},i}}{k_4+k_1}.$$

Here, k_2 is the number of solutions in the archive dominating some solutions in new-pts, k_3 is the number of solutions in new-pts dominated by some solutions in the archive, and k_4 is the number of such domination. Note that probability calculation considers more exploration around new solution at higher temperature, which is typical for a simulated annealing algorithm [24]

2) *Case 2:* Since all solutions in new-pts are nondominating w.r.t. cur-pt, we randomly pick one solution and assign it to cur-pt based on the probability in (2) with $k_1 = 0$.
3) *Case 3:* Since there are solutions in new-pts dominating cur-pt, one of them is chosen randomly and assigned to cur-pt.

Next, for updating the archive, we use solutions in new-pts to replace dominated solutions in the archive. By merging several cases/subcases from the original algorithm [22] into just three cases, we have improved the understanding of the algorithm and made the pseudocode easier to implement. Additionally, while the original algorithm incorporates clustering to mitigate the loss of diversity in solutions, we have chosen not to employ clustering due to the relatively small size of the solution set in our problem. A pseudocode for our CC placement optimization algorithm is presented in Algorithm 1.

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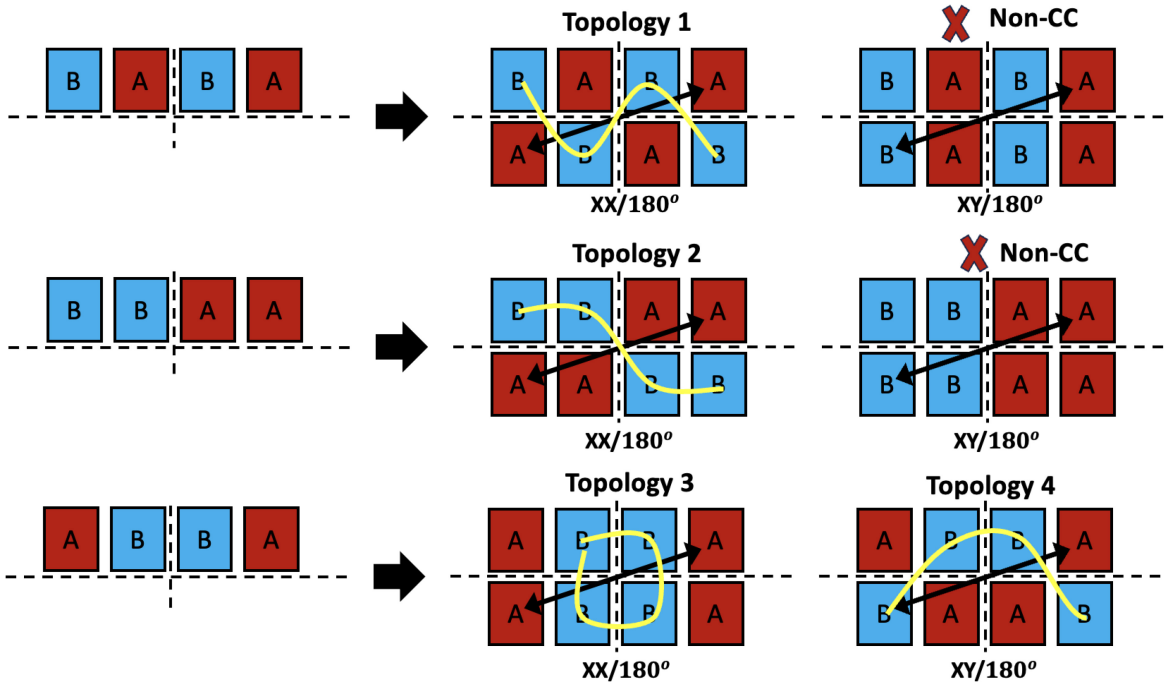


Fig. 3. Applying XX/180° and XY/180° (where, X, Y ∈ {A, B, C, D}, X ≠ Y) transformations on a two transistors schematic with four units each. Out of the six layouts produced by the transformation four are CC type.

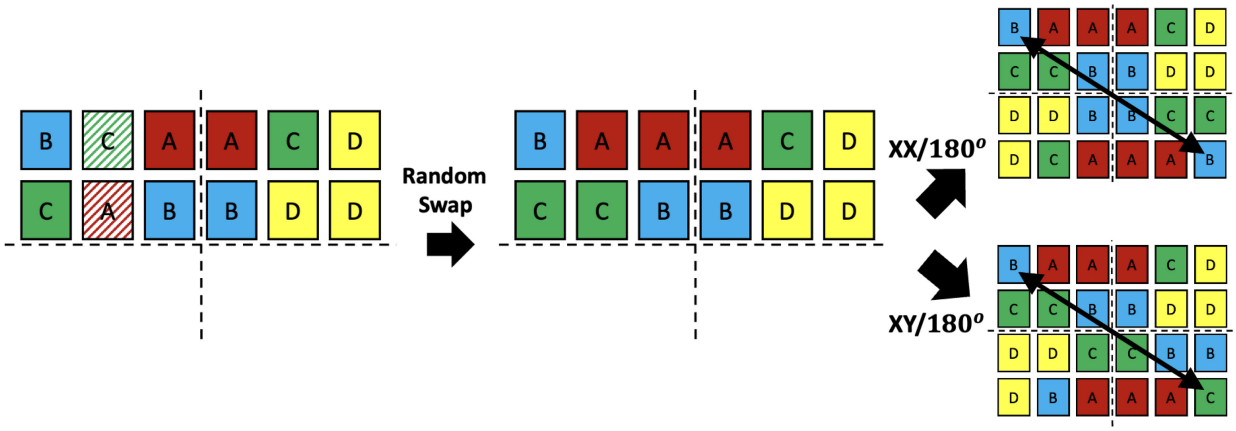


Fig. 4. Random swap (A <=> C) and XX/180° and XY/180° transformations for a more general case of perturbation. XY/180° transformation yields CC topology only when applied on devices B and C.

III. Layout Constraints

Next we discuss the different layout constraints, i.e., the objectives functions (f_i for $i = 1$ to M in (1)) to be handled by the proposed optimization algorithm.

A. Degree of Dispersion

One of the fundamental rules for CC layouts underscores the importance of achieving the maximum degree of dispersion [1]. This involves distributing the device uniformly throughout the array. In [2], a quantitative measure of the degree of dispersion has been proposed. Assume there is an edge between adjacent units in a placement solution. Then, the following expression captures the degree of dispersion:

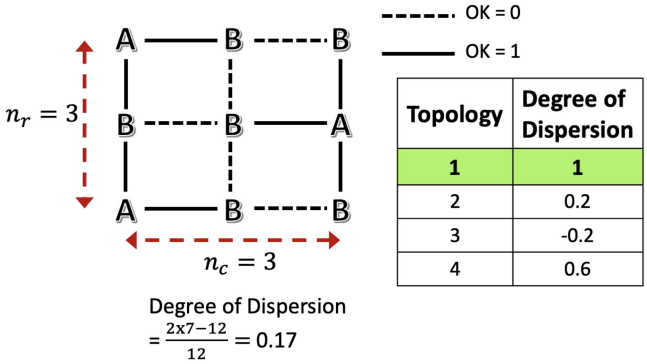
$$\frac{2\sum \text{OK} - 2n_c n_r + (n_c + n_r)}{2n_c n_r - (n_c + n_r)}. \tag{3}$$

Here, if an edge connects units belonging to different devices, OK is 1, otherwise OK is 0. n_c and n_r are, respectively, the number of columns and rows in a CC placement. This is illustrated in Fig. 5. The Degree of Dispersion, measured within the range $(-1, 1]$, signifies a higher value as being better. A higher value indicates increased spreading of the device units throughout the layout, resulting in reduced clustering. In Fig. 5, in the Table we show the degree of dispersion values for the four topologies. Among the four, topology 1 has the highest dispersion value, reflecting a uniform spread of both transistors along both the X and Y axes. Topology 3 exhibits the lowest dispersion, attributed to the clustering of B units along both the X and Y axes. Comparing topologies 4 and 2, the latter exhibits more clustering along the X axis. When comparing topologies 1 and 4, the former exhibits a better interdigitation along the X-axis.

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Algorithm 1 CC Placement

Set $T_{\max}$, $T_{\min}$, *iter*, α, $temp = T_{\max}$
Get cur-pt with minimum diffusion break and add it to the archive
while $temp > T_{\min}$ **do**
  for $i = 0$; $i <$ *iter*; $i{+}{+}$ **do**
    Perturb cur-pt to get new-pts /* Random swap and transformation (XX/180°, XY/180°) */
    Remove non-CC solutions from new-pts
    Remove solutions in new-pts having diffusion break and dummy count above upper bounds
    Keep only nondominated solutions in new-pts
    // Assume size of new-pts is now k
    /* update cur-pt */
    if cur-pt dominates k_1 (> 0) solutions in new-pts **then** /* **Case 1** */
      Randomly pick a solution new-pt from $k{-}k_1$ solutions
      Assign new-pt to cur-pt with probability=*prob* (2)
    end if
    if cur-pt and new-pts are nondominating to each other **then** /* **Case 2** */
      Randomly pick a solution new-pt from new-pts
      Assign new-pt to cur-pt with probability=*prob* (Eq. (2) with $k_1 = 0$)
    end if
    if k_1 (> 0) solutions in new-pts dominate cur-pt **then** /* **Case 3** */
      Randomly pick a solution new-pt from k_1 solutions
      Assign new-pt to cur-pt
    end if
    /* update archive */
    Replace the dominated solutions in archive with new-pts
  end for
  $temp = \alpha * temp$
end while
Get the desired solution from the archive



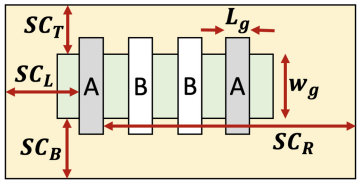
Topology	Degree of Dispersion
1	1
2	0.2
3	-0.2
4	0.6

Fig. 5. Defining degree of dispersion for devices A and B having 3 and 6 units, respectively. Degree of Dispersion values for four layouts from Fig. 3. Topology 1 has the highest degree of dispersion.

B. Layout Dependent Effects

We account for the WPE as it is a critical LDEs [29], [30]. WPE captures variations in the threshold voltage based on the distance of the transistor to the well-edge, as illustrated in Fig. 6

$$\Delta V_{\text{th}} \propto \frac{1}{\text{WPE}} = \sum_{i=1}^{n}\left(\frac{1}{\text{SC}_L^i + L_g} + \frac{1}{\text{SC}_R^i + L_g} + \frac{1}{\text{SC}_T^i + W_g} + \frac{1}{\text{SC}_B^i + W_g}\right). \quad (4)$$



Topology	LDE
1	0
2	0
3	0.21
4	0

Fig. 6. WPE. The Table shows LDEs that capture LOD and WPE for the four topologies from the Fig. 3.

Here, n represents the number of unit cells, while L_g and W_g denote the gate length and width of a unit, respectively. SC_L, SC_R, SC_T, and SC_B are the distances from the left, right, top, and bottom well boundaries. To mitigate the threshold voltage variations (ΔV_{th}) introduced by the WPE, we minimize the difference, or mismatch, of the mean values of (1 WPE) across all devices, captured by the following model:

$$\sum_{k=1}^{N-1}\sum_{l=k+1}^{N}\left|\frac{\left(\frac{1}{\text{WPE}}\right)_k}{n_k} - \frac{\left(\frac{1}{\text{WPE}}\right)_l}{n_l}\right|. \quad (5)$$

Here, N is the number of devices and n_i is the number of unit cells of the device i. $(1/\text{WPE})_i$ is the (1/WPE) value of the device i. The following simplification of WPE is sufficient for comparison purpose:

$$\left(\frac{1}{\text{WPE}}\right)_i = \sum_{u=1}^{n_i}\frac{1}{x_u} + \frac{1}{r+1-x_u} + \frac{1}{y_u} + \frac{1}{c+1-y_u}. \quad (6)$$

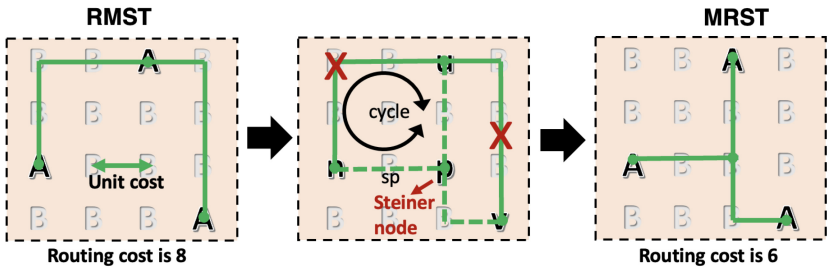
x_u and y_u are, respectively, the column and row of the unit cell u, r and c are the number of unit cells in each row and column, respectively. Note that the first two terms also capture the LOD [19], [21]. For the four layouts in Fig. 3 the LDE values are shown in the Table in Fig. 6. Except topology 3, all other topologies have LDE value 0.

C. Routing Cost

In their work, [21] utilized the concept of rectilinear minimum spanning tree (RMST) to calculate routing cost. However, RMST can lead to overlapping edges, potentially costing up to 1.5 times longer wire length than minimum rectilinear steiner tree (MRST) [31]. Although calculating the steiner tree is an NP-complete problem, many fast heuritics have been proposed. We use one of the very first algorithm for steiner tree calculation proposed in [31] due to ease of implementation and suitability for the problem size of ours.

The algorithm begins by constructing an undirected graph, where each unit cell connected to the net is represented as a node. Rectilinear edges connect every pair of nodes on the net, and the RMST is obtained using Prim's algorithm. Subsequently, for each node (n) and edge pair (u, v) in the RMST, a new steiner node that improves the wirelength is obtained. The shortest path (sp) from the node (n) to the edge (u, v) or the rectangular layout of the edge is then calculated. The newly introduced node (p) on the edge or the rectangle becomes the steiner node. The edge (u, v) is replaced by the

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Topology	Routing cost	
1	RMST	36
	MRST	30
2	RMST	24
	MRST	24
3	RMST	24
	MRST	24
4	RMST	30
	MRST	30

Fig. 7. Routing cost calculation for device A having 3 units. Topologies 2 and 3 have the minimum routing cost. MRST yields smaller routing cost for the topology 1.

Algorithm 2 trialAddSteiner

```
1: trialAddSteiner (G, n, (u, v))
2: find shortest distance node p on rectangular layout of (u, v) from
   n /* p is the steiner node */
3: G.node ← p
4: remove (u, v) from G
5: G.edge ← (p, u)
6: G.weight ← |loc.p.x - loc.u.x| + |loc.p.y - loc.u.y|
7: G.edge ← (p, v)
8: G.weight ← |loc.p.x - loc.v.x| + |loc.p.y - loc.v.y|
9: G.edge ← (n, p)
10: G.weight ← |loc.n.x - loc.p.x| + |loc.n.y - loc.p.y|
11: find the edge with largest weight (u, n) in G
12: remove (u, n) from G
13: gain = weight(u, n) - weight(n, p)
14: return <gain, G>
15: end trialAddSteiner
```

Algorithm 3 Calculate Routing Cost

```
Input: topology, netlist
1: routing_cost = 0
2: for each net in netlist do
3:     /* Construct the graph */
4:     for each unit connected to net do
5:         G.node ← unit
6:     end for
7:     for each pair of node (u, v) in G do
8:         G.edge ← (u, v)
9:         find location (loc) of u and v in topology
10:        G.weight ← |loc.u.x - loc.v.x| + |loc.u.y - loc.v.y|
11:    end for
12:    /* Find Rectilinear Minimum Spanning Tree (RMST) */
13:    find G_RMST in G using Prim's algorithm
14:    /* Find Minimum Rectilinear Steiner Tree (MRST) */
15:    while there is improvement in sum(G_RMST.weight) do
16:        i = 1
17:        for each <n, (u, v)> in G_RMST do
18:            <Gain[i++], tmp> = trialAddSteiner (G_RMST, n, (u, v))
19:        end for
20:        find max(Gain) and corresponding <n, (u, v)>
21:        <tmp, G_RMST> = trialAddSteiner (G_RMST, n, (u, v))
22:    end while
23:    G_MRST = G_RMST
24:    routing_cost = routing_cost + sum(G_MRST.weight)
25: end for
26: return routing_cost
```

edges (p, u), (p, v), and (n, p), forming a cycle. The algorithm iteratively identifies and removes the edge with the largest weight (u, n) until no further improvement is observed. For the example in Fig. 7, removing (u, n) from the cycle n->p->u->n results in a gain of weight(u, n) − weight$(n, p) = 4 - 2 = 2$.

The routing costs for the four layouts from the Fig. 3 are presented in Fig. 7. Topologies 2 and 3 have the best routing cost due to more clustering of units A and B compared to other topologies. The tradeoff between the routing cost and the degree of dispersion can be noted. MRST yields smaller routing costs than RMST for the topology 1. A pseudocode for the routing cost calculation is presented in Algorithm 3, which calls Algorithm 2.

IV. Results and Discussion

The proposed algorithm has been implemented in C++, and all experiments have been conducted on a Linux environment with Intel Core CPU running at 3.3 GHz with 128 GB of memory. In Algorithm 1, we set $T_{\max}$, $T_{\min}$, α, and *iter* as 100, 10^{-7}, 0.37 and 100, respectively. In the optimization formulation, we set the degree of dispersion, layout-dependent effects (LDEs) and routing cost as the objective functions. Our goal is to maximize the degree of dispersion and minimize LDE and routing cost. Diffusion break ($C_{\text{diff_break}}$) and dummy count ($C_{\text{dummy_count}}$) values from state-of-the-art solutions are used as constraints. Therefore, our optimization involves finding a placement solution sol = sol* that

$$\begin{aligned}\text{minimizes}\ \ &\{1/f_{\text{degree of dispersion}}(\text{sol}),\\ &f_{\text{LDE}}(\text{sol}),\ f_{\text{routing cost}}(\text{sol})\}\end{aligned}$$

while satisfying

$$\begin{aligned}f_{\text{diffusion break}}(\text{sol}) &\le C_{\text{diff_break}}\\ f_{\text{dummy count}}(\text{sol}) &\le C_{\text{dummy_count}}.\end{aligned}$$

To add a new objective, we need a model (i.e., f) for the new objective, which takes placement solution (*sol*) as the input and outputs the value (f(sol)) to be optimized (refer to (1) and (2)). We have used TSMC 40 nm PDK in Cadence Virtuoso schematic and GXL layout environment (version IC6.1.8-64b.500.17) [32]. For placement, we manually create

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TABLE I
COMPARISON WITH [19] FOR FIVE CM CONFIGURATIONS REPORTED IN [19]

Test case	Method	Degree of Dispersion	LDE	Mismatch in Current	Routing Cost		Diffusion Break	Dummy Count	Layout Area (μm^2)
					Model	Router			
CM:1	[19]	0.37	0.50	2.38	75	78	0	0	15
[2,2,4,8,8], K=1.3	This work	0.47	0.21	3.39	77	79	0	0	15
CM:2	[19]	0.04	0.59	3.1	55	63	0	0	12
[2,2,4,10], K=2	This work	0.19	0.46	0.7	55	63	0	0	12
CM:3	[19]	0.17	0.47	1.8	46	52	0	0	10
[2,2,4,8], K=1.3	This work	0.17	0.39	1.7	46	51	0	0	10
CM:4	[19]	0.58	0.58	3.72	76	74	0	0	14
[4,4,8,8], K=1.3	This work	0.26	0.34	0.1	69	71	0	0	14
CM:5	[19]	0.38	0.73	4.46	108	112	0	0	22
[4,4,4,10,10], K=2	This work	0.23	0.57	1.02	99	110	0	0	22

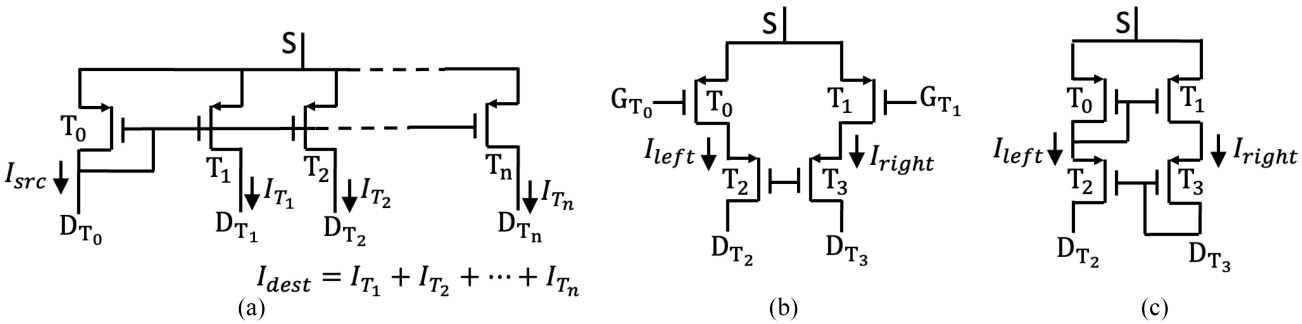


Fig. 8. (a) CM. (b) CDIP. (c) CDLP.

the topology (i.e., the pattern) using the Modgen feature and autoroute it. Labels and NWELL are created manually, and we perform post-layout extraction using Calibre nmLVS and PEX (version 2023.4_17.10), which are integrated into the Virtuoso platform. We run DC simulations using Spectre through ocean script.

To benchmark our approach against state-of-the-art solutions [19], we have used five configurations of the current mirror structure (CM:1-5), from [19]. The results for the five test cases are shown in Table I. Both algorithms generate solutions without any diffusion break. There is a good improvement in the routing cost, as predicted by both the routing model and the auto-router. For the current mirror (CM) circuit, we use mismatch expression $100|n\ I_{src} - I_{\text{dest}}|/(n\ I_{src} + I_{\text{dest}})$, where n is the CM ratio (e.g., for CM:1, $n = 11$), I_{src} is the current flowing through the source transistor [e.g., D_{T_0} in Fig. 8(a)] and I_{dest} is the total current flowing through the destination transistors [e.g., D_{T_1} to D_{T_n} in Fig. 8(a)] which copy current from the source transistor. Ideally, the value should be 0 when the source and destination transistors match. With the exception of CM:1, in all test cases, there is an improvement in current mismatch value. It is important to note that the degree of dispersion, the LDEs and the parasitics all play a role in current matching performance. A smaller degree of dispersion suggests less uniform device spread, potentially degrading spatial variation performance. Parasitic resistance contributes to IR drop, thereby affecting transistor current, LDEs affect transistor current by causing change in the threshold voltage. We do not consider random mismatch (such analysis is usually done by monte-carlo simulation) in our simulation, as CC layout is not useful for canceling random variation.

We have created an additional set of six tests, incorporating two scenarios from each of the three distinct configurations: CM, Cascode differential input pair (CDIP), and Cascode differential load pair (CDLP) as shown in Fig. 8. The results for these test cases are shown in Table II. In half of the test cases, improvement in diffusion break or dummies is observed. This is particularly significant as a reduction in dummies means that the area usage is smaller, which can reduce routing cost, and the impact of variation. Note that our algorithm optimizes both the dummy count and the diffusion break. We have observed that having only dummy count as the optimization objective gives inferior results. This could be attributed to the diffusion break having a smaller search space, which helps the optimizer avoid getting stuck at local minima. Fig. 9 shows how dummy count can increase when no constraint is set on the diffusion break. In our experiment, we have used the state-of-the-art solution as a constraint to limit the dummy count and the diffusion break. However, the initial placement solution can also be set as a constraint.

Across different circuit configurations, there is a good improvement in routed wirelength. For measuring the current mismatch in CDIP and CDLP, we use the expression $100|I_{\text{left}} - I_{\text{right}}|/(I_{\text{left}} + I_{\text{right}})$, representing the difference in current between the left and right branches of the differential pair with the resultant difference normalized by the total current. Both Tables have more instances of green than red, signifying good improvement. It is important to underscore that routing cost, mismatch in current, and layout area serve as practical metrics for improvement. Across various circuit configurations, our algorithm performs better than state-of-the-art algorithm in

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TABLE II
COMPARISON WITH [19] FOR SIX RANDOM TESTCASES: CM, CDIP, AND CDLP

Test case	Method	Degree of Dispersion	LDE	Mismatch in Current	Routing Cost		Diffusion Break	Dummy Count	Layout Area (μm^2)
					Model	Router			
CM:1 [2,2,2,2,10], K=2	[19]	0.19	0.97	2.04	56	78	2	6	15
	This work	0.33	0.92	1.06	53	73	2	6	15
CM:2 [2,2,2,6,6], K=2	[19]	0.48	0.98	3.46	58	75	2	6	16
	This work	0.48	0.82	3.11	53	72	2	6	16
CDIP:1 [6,6,10,10], K=2	[19]	0.54	0.21	1.05	124	186	4	8	28
	This work	0.69	0.01	0.02	122	139	0	0	24
CDIP:2 [10,10,10,10], K=2	[19]	0.58	0.10	0.01	150	227	4	16	35
	This work	0.58	0.07	0.50	146	194	2	8	30
CDLP:1 [2,2,6,6], K=2	[19]	0.33	0.56	0.10	47	66	0	0	12
	This work	0.17	0.56	4.49	43	57	0	0	12
CDLP:2 [6,6,6,6], K=1.3	[19]	0.58	0.42	6.90	64	100	4	12	20
	This work	0.58	0.22	0.37	61	87	2	4	16

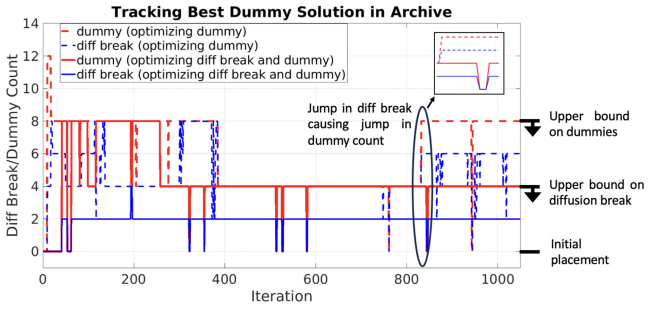


Fig. 9. Shows how limiting diffusion break to not go above the upper bound helps keep dummy count in check.

TABLE III
COMPARING THE NUMBER OF SOLUTIONS PRODUCED BY AMOSA [22] AND THIS WORK

Test case	Method	No. of sols	No. of XY sols
CM:1, [2,2,2,2], K=2	AMOSA [22]	12	0
	This work	12	0
CM:2, [4,4,4,4], K=2	AMOSA [22]	141	73
	This work	147	78
CM:3, [6,6,6,6], K=2	AMOSA [22]	49	11
	This work	71	24
CM:4, [8,8,8,8], K=2	AMOSA [22]	66	7
	This work	50	11
CM:5, [10,10,10,10], K=2	AMOSA [22]	50	5
	This work	66	4

these parameters. The runtime of our algorithm is about 1 min for the largest test case, CDIP:2.

Note that we have expressed the current difference as a percentage of the total bias current for all the circuits as it is a more meaningful measure of mismatch than the absolute current difference. A smaller absolute current difference does not necessarily indicate a well-matched circuit or better mismatch performance. In circuit with very low bias current, even a smaller current difference can indicate worse mismatch performance. Thus, the normalized current difference we reported is a better metric for comparing relative performance across different circuits.

A. Multiobjective Optimization

Multiobjective optimization algorithm may produce numerous nondominant solutions, necessitating the use of clustering techniques to reduce the number of solutions [22]. However, in the problem handled in this article, we observe fewer than a hundred solutions, owing to the relatively smaller search space, especially considering integer nature of the diffusion break or dummy solution. We therefore do not employ the clustering technique. To obtain the optimized solution presented in Tables I and II from the set of solutions, we have done some post-processing, assigning greater weight to solutions having better diffusion break or dummies, routing cost, and LDEs. These selected solutions usually have worse degree of dispersion value, showing competing nature of the different performance metrics. There are solutions with a better degree of dispersion compared to state-of-the-art, however, such solutions have degradation in other critical parameters. Highlighting the distinction from single-objective optimization, where we indirectly influence the final solution by adjusting the objectives' coefficients, in the current approach, we have a set of solutions from which we select the best solution based on the target requirements. This is an important advantage over state-of-the-art methods, as they cannot produce competing or nondominating solutions.

1) Comparison to AMOSA ("One Solution" Approach): We have discussed with the authors of the AMOSA [22] paper about the enhancement proposed in this article. The direct comparison with AMOSA results from that discussion. Note that AMOSA can be modified in a straightforward way to handle multiple solutions in each iteration of the optimization by randomly selecting one solution. However, we have observed that our approach to handling multiple solutions yields better quality solutions. For comparison purposes, we have reported in Table III the number of nondominating solutions produced by the algorithms using purity measure [22].

The purity measure calculates the fraction of solutions within the range [0, 1] produced by a specific algorithm that remains nondominating after merging the final solutions from all algorithms. However, a high purity fraction may still correspond to a small number of solutions. Since an algorithm that generates more solutions provides users with more design options, this quantity is a more meaningful indicator of the algorithm's quality for our problem. Therefore, instead of using fractions, we report the number of solutions

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TABLE IV
COMPARING THE NUMBER OF SOLUTIONS PRODUCED BY THE ALGORITHMS CONSIDERING DIFFERENT TEMPERATURE EFFECT IN PROBABILITY CALCULATION

Test case	Method	Baseline: AMOSA [22]		Baseline: This work	
		No. of sols	No. of XY sols	No. of sols	No. of XY sols
CM:1, [2,2,2,2], K=2	Eq. (2) in AMOSA [22]	12	0	12	0
	Eq. (2)	12	0	12	0
CM:2, [4,4,4,4], K=2	Eq. (2) in AMOSA [22]	141	73	149	78
	Eq. (2)	144	74	147	77
CM:3, [6,6,6,6], K=2	Eq. (2) in AMOSA [22]	62	17	59	20
	Eq. (2)	69	5	57	18
CM:4, [8,8,8,8], K=2	Eq. (2) in AMOSA [22]	45	2	47	12
	Eq. (2)	70	6	80	21
CM:5, [10,10,10,10], K=2	Eq. (2) in AMOSA [22]	53	2	37	3
	Eq. (2)	82	1	82	7

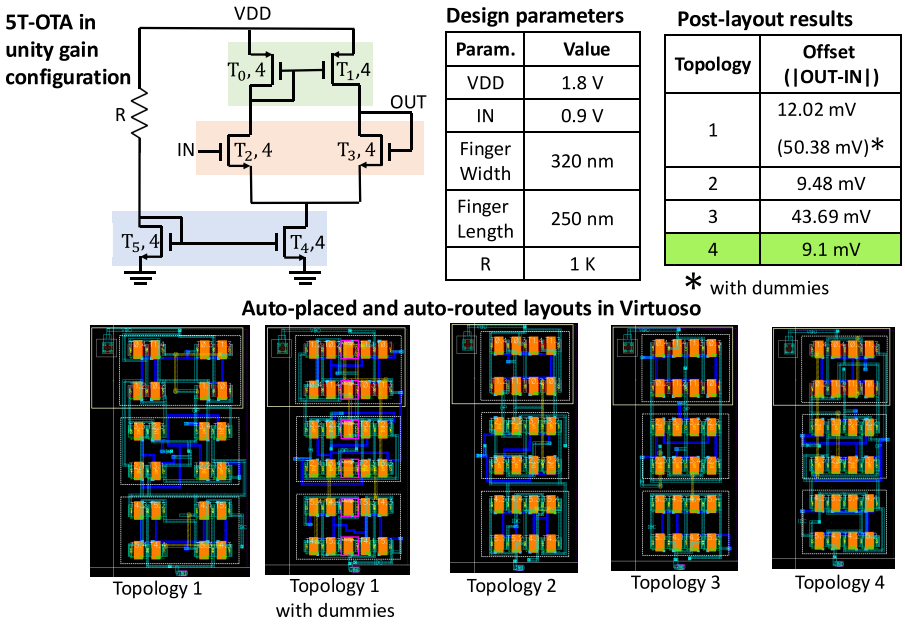


Fig. 10. 5T-OTA shows best offset performance with CC topology 4. Auto-placed and auto-generated layouts for all the topologies are shown in the figure. Resistor (R) is part of the testbench and not included in the layout.

after such merging. This comparison does not include the new temperature effect in (2) proposed in this article. Our algorithm not only produces more solutions, but the solution space is more diverse as it has more XY solutions despite the number of XY solutions being generally fewer than XX solutions. Note that the XY search space is more than XX search space, however, most of the XY solutions do not qualify as CC. For this experiment, we have used five CM test cases that are different from the CM example presented in Tables I and II. These test cases have a more diverse search space spread over XX and XY compared to the previously presented test cases.

Regarding the probability calculation, the inverse effect of temperature in (2) in AMOSA [22] generally yields inferior results compared to (2), as shown in Table IV. For this experiment, we have used AMOSA [22] and the proposed algorithm as the baseline. When AMOSA is used as the baseline, the new temperature effect in (2) (i.e., the probability being proportional to temperature) produces more solutions than (2) in AMOSA [22] (i.e., the probability being inversely related to temperature) in more test cases. When using the proposed algorithm as the baseline, the degradation is minimal for test cases CM:2 and CM:3, while the improvements in test cases CM:4 and CM:5 are significant. Therefore, the overall performance is better. When the new temperature effect is considered together with the multisolutions approach (i.e., the proposed algorithm), the results are better than the AMOSA (follow Baseline: AMOSA and Baseline: This work with (2)).

B. Post-Layout Simulation Results of 5T-OTA

We present post-layout simulation results of a 5T-OTA circuit in Fig. 10. Each transistor in the OTA comprises four units with design parameters listed in the Table. They are arranged in groups ($<T_0\ T_1>$, $<T_2\ T_3>$, $<T_4\ T_5>$), and laid out following the four CC topologies as illustrated before in Fig. 3. The layout is done using the same Cadence Virtuoso auto-placer and auto-router [32]. To measure the offset voltage, the OTA is connected in unity gain configuration.

A schematic level simulation of the OTA yields an offset voltage of 3.41mV. In post-layout, there is degradation in the offset as reported in the Table in Fig. 10. Note that except topology 1, all other topologies can fully share diffusion region without using dummies. Topology 1 has 12.02 mV and 50.38 mV offset, respectively, without and with dummies. The

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offset for topology 1 increases when dummy transistors are added to share the diffusion region, likely due to added LDE and parasitic effects. As shown in Figs. 5 and 6, topology 3 has the lowest degree of dispersion and the highest LDE among all topologies, which may explain its large variation in offset performance. Although topology 3 has the best routing cost (Fig. 7), routing plays a less significant role at low frequency as the routing length is usually short at the device level. Intuitively, this makes sense, as topology 3 clusters all units of the B device together, and the units from device A are placed along the edges on both sides, leading to significant mismatch. While topology 4 yields the best result, it is not always the ideal layout for all three groups in the OTA. The optimal result will likely come from employing different topologies for each group. This, however, is a circuit-level optimization problem where we have to also consider the impact of circuit block placement and routing on the final circuit performance—this is beyond the scope of the discussion here. Our point is that without exploring diverse topologies at device-level placement, achieving the best circuit performance is not possible. We have addressed the device-level placement issue in this article.

V. Conclusion

We have introduced a unified multiobjective optimization framework for CC placement of analog transistors. We enhance a well-known multiobjective optimization algorithm, AMOSA, to handle more than one new solution per iteration. Moreover, we consolidate several cases/subcases of the AMOSA algorithm into just three cases, simplifying the algorithm and making the code easier to implement. Our formulation includes important layout constraints. The proposed method enables exploration of significantly more topologies by applying powerful transformations. In contrast to existing methods, our approach shows better performance in post-layout simulation, consistently generating more optimal CC placements across diverse circuit configurations.

It is important to note that CC layouts are generally not preferred for high-frequency applications due to high parasitic effects. In future studies, we plan to optimize CC structures considering high-frequency metrics. This would involve developing a routing infrastructure for symmetric routing and focusing on minimizing routing mismatch, rather than minimizing absolute parasitics.

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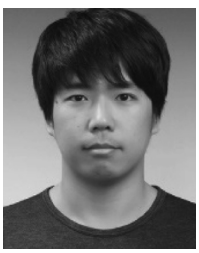
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